

NrgOpt — Photovoltaic Project Site Survey Checklist / 光伏项目现场踏勘表 / 太陽光発電プロジェクト現地調査票

Project Information | 项目信息 | プロジェクト情報

Item / 项目 / 項目	Detail / 内容 / 内容
Project / Owner Name	项目名称/业主名称
Project Address	项目地址
GPS Coordinates (lat/lng)	经纬度坐标
Owner Contact Person	业主联系人
Owner Contact Phone	业主联系电话
Survey Date	踏勘日期
Survey Engineer	踏勘工程师
Weather During Survey	踏勘时天气

1. Building Specifications | 建筑参数 | 建物仕様

1.1 Basic Dimensions | 基本尺寸 | 基本寸法

- ☐ Building area (m²): _
- ☐ Elevation (m): _
- ☐ Number of floors: _
- ☐ Building orientation (sun-facing direction): _
- ☐ Year built / time in use: _
- ☐ Site usage / ownership type: _

1.2 Structural Load | 结构承载力 | 構造耐力

- ☐ Roof design live load (kg/m²): _
- ☐ Current load status (existing equipment weight): _
- ☐ Remaining capacity (kg/m²): _
- ☐ Structural reinforcement required? ☐ Yes ☐ No
- ☐ Building seismic rating: _

- [] Structural inspection report available? ☐ Yes ☐ No

1.3 Roof Type | 屋顶类型 | 屋根タイプ

Concrete roof (水泥楼面 / コンクリート屋根):

- [] Length × Width × Height (m): _
- [] Parapet wall height (m): _
- [] Staircase / opening locations and dimensions: _
- [] Roof pitch / slope: _
- [] Water leakage condition (describe): _
- [] Any repairs, contamination, or equipment on roof? _
- [] Skylight area / light strips: _
- [] Obstacle dimensions (L×W×H): _
- [] Roof waterproofing condition: _

Metal sheet roof (彩钢瓦 / 金属屋根):

- [] Sheet profile type: ☐ Standing seam ☐ Angle brace ☐ Trapezoidal
- [] Roof pitch: _
- [] Beam specification: _
- [] Beam span (m): _
- [] Purlin specification: _
- [] Purlin span (m): _
- [] Column specification: _
- [] Purlin rust condition: _
- [] Roof surface rust condition: _
- [] Roof panel thickness (mm): _
- [] Does drawing match actual roof? ☐ Yes ☐ No

1.4 Obstruction Survey | 遮挡物调查 | 障害物調査

- [] Obstruction 1: L_ × W_ × H (m), **Direction from roof centre:** ____
- [] Obstruction 2: L_ × W_ × H (m), **Direction from roof centre:** ____
- [] Obstruction 3: L_ × W_ × H (m), **Direction from roof centre:** ____
- [] Additional notes: _____

1.5 Shading Analysis | 阴影分析 | 日影分析

- [] Surrounding buildings (height + distance from site):
- Building 1: height__m, distance_m, direction__
- Building 2: height__m, distance_m, direction__
- [] Trees or vegetation casting shadows: ☐ Yes ☐ No (describe: ____)
- [] Chimneys / exhaust pipes nearby: ☐ Yes ☐ No (height: __m, distance: ____m)
- [] Shading from adjacent roof structures (e.g., elevator room, HVAC): ☐ Yes ☐ No
- [] Winter solstice sun path checked? ☐ Yes ☐ No
- [] Drone shadow simulation performed? ☐ Yes ☐ No

1.6 Solar Panel Layout | 组件排布 | パネル配置

- [] Rail spacing (mm): _
 - [] Tilt angle planned (°): _
 - [] Row spacing (m): _
 - [] Panel orientation: ☐ Landscape ☐ Portrait
 - [] Estimated number of panels: _
 - [] PV power station height (m): _
-

2. Meteorological & Environmental | 气象与环境 | 気象・環境

2.1 Climate Data | 气象数据 | 気象データ

- [] Annual solar irradiation (kWh/m²): _
- [] Maximum wind speed (m/s): _
- [] Snow load (kN/m²): _
- [] Extreme low temperature (°C): _
- [] Extreme high temperature (°C): _
- [] Annual average temperature (°C): _
- [] Annual rainfall (mm): _
- [] Lightning frequency: ☐ Low ☐ Medium ☐ High

2.2 Environmental Risks | 环境风险 | 環境リスク

- [] Site within 5km of coastline? ☐ Yes ☐ No (salt spray risk)
 - [] Nearby chemical / cement / steel plants? ☐ Yes ☐ No (dust/particulate risk)
 - [] Nearby water bodies (flood risk)? ☐ Yes ☐ No
 - [] Site in earthquake-prone zone? ☐ Yes ☐ No
 - [] Site in typhoon path? ☐ Yes ☐ No
 - [] Soil corrosion level: ☐ Low ☐ Medium ☐ High
 - [] Bird activity level: ☐ Low ☐ Medium ☐ High (soiling risk)
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3. Electrical Infrastructure | 电气设施 | 電気設備

3.1 Grid Connection | 并网接入 | 系統連系

- [] **Grid connection type:**
 - ☐ Low-voltage (LV) grid connection
 - ☐ High-voltage (HV) grid connection
- [] Number of transformers: _
- [] Grid-connected transformer capacity (kVA): _
- [] Transformer model / winding group: _
- [] Existing protection devices (model): _

- [] Any existing solar PV system? ☐ Yes (capacity: _____kW) ☐ No
- [] Distance from roof to grid connection point (m): _

3.2 Distribution Room | 配电房 | 配電室

- [] Distribution room location / photo taken: ☐ Yes ☐ No
- [] Is there space for PV grid connection cabinet? ☐ Yes ☐ No
- [] Spare breaker slots available? ☐ Yes (how many: ____) ☐ No
- [] Cable trench available? ☐ Yes ☐ No
- [] AC cable requires road crossing? ☐ Yes ☐ No
- [] Single-line diagram available? ☐ Yes (attached) ☐ No
- [] Transformer room temperature / ventilation: _

3.3 Consumption Profile | 用电负荷 | 電力負荷

- [] **Power generation mode:**
- ☐ Full grid export (全额上网)
- ☐ Self-consumption with surplus export (自发自用，余额上网)
- [] Contracted demand (kW): _
- [] Hourly load profile available? ☐ Yes (attach) ☐ No
- [] Peak load period: _ :00 – _ :00
- [] Weekend / holiday operation? ☐ Yes ☐ No
- [] Night shift operation? ☐ Yes ☐ No
- [] Seasonal load variation? _

3.4 Monthly Consumption | 月用电量 | 月間電力消費

Month	kWh	Month	kWh
Jan		Jul	
Feb		Aug	
Mar		Sep	
Apr		Oct	
May		Nov	
Jun		Dec	

Annual total: _ kWh

3.5 Electricity Tariff | 电价信息 | 電気料金

- [] Electricity supply company: _
- [] Tariff category: ☐ Industrial ☐ Commercial ☐ Other
- [] Peak period rate (元/kWh): _
- [] Flat period rate (元/kWh): _
- [] Off-peak period rate (元/kWh): _

- [] Average weighted rate (元/kWh): _
 - [] Demand charge (元/kW/month): _
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4. Site Access & Logistics | 进场与施工 | アクセスと施工

4.1 Access Assessment | 进场条件 | アクセス条件

- [] Road width to site entrance (m): _
- [] Truck access (max vehicle length): _
- [] Crane working radius (m): _
- [] Crane parking position identified? ☐ Yes ☐ No
- [] Road closure / traffic permit required? ☐ Yes ☐ No
- [] Material staging area available? ☐ Yes (size: _____m²) ☐ No
- [] Elevator / hoist for material lifting? ☐ Yes ☐ No (describe alternative: _)
- [] Construction power supply available? ☐ Yes ☐ No
- [] Construction water supply available? ☐ Yes ☐ No

4.2 Safety Considerations | 安全考量 | 安全事項

- [] Roof edge protection / guardrails present? ☐ Yes ☐ No
 - [] Roof access (stairs / ladder / hatch): _
 - [] Fall protection anchor points? ☐ Yes ☐ No
 - [] Emergency evacuation route identified? ☐ Yes ☐ No
 - [] Asbestos or hazardous materials present? ☐ Yes ☐ No
-

5. Regulatory & Documentation | 合规与资料 | 規制・書類

5.1 Owner Documentation | 业主文件 | 所有者書類

- [] Property ownership certificate (房屋产权证)
- [] Land use certificate (土地使用证)
- [] Building construction drawings (房屋建造图纸)
- [] Structural calculation report (结构计算书)
- [] Electrical single-line diagram (电气一次图)
- [] Fire safety approval (消防验收)
- [] Environmental impact assessment (环评)
- [] Business license (营业执照)
- [] CAD files
- [] PDF documents

5.2 Regulatory Check | 合规检查 | 規制確認

- [] Local distributed PV filing requirements: _
- [] Grid company pre-approval required? ☐ Yes ☐ No
- [] Building height restriction applicable? ☐ Yes ☐ No
- [] Heritage / conservation area? ☐ Yes ☐ No
- [] Aviation restriction applicable? ☐ Yes ☐ No
- [] Neighbour consent required? ☐ Yes ☐ No

6. Photo Checklist | 现场照片清单 | 写真チェックリスト

#	Shot Description	Taken (✓)
1	North-facing view from roof	
2	East-facing view from roof	
3	South-facing view from roof	
4	West-facing view from roof	
5	Roof panorama (drone / wide angle)	
6	Roof surface close-up × 4 corners	
7	Obstructions close-up (each)	
8	Parapet wall / roof edge detail	
9	Existing equipment on roof	
10	Distribution room — exterior	
11	Distribution room — interior (full panel view)	
12	Distribution room — spare breaker slots	
13	Transformer nameplate	
14	Cable route from roof to distribution room	
15	Site entrance and access road	
16	Surrounding buildings from ground level	
17	Roof access point (stairs / ladder / hatch)	
18	Drone video (full building + surroundings)	
19	GPS-tagged location photo	
20	Team photo at site	

7. Cost & Timeline Estimation | 成本与工期 | コストと工期

- [] Estimated cable length from roof to grid connection (m): _
- [] Estimated cable trenching required (m): _
- [] Special access equipment required? ☐ Yes (specify: _) ☐ No
- [] Grid outage window available? ☐ Yes (dates: _) ☐ No
- [] Expected construction start: _
- [] Expected construction duration (days): _
- [] Any seasonal constraint? _

8. Sign-off | 签字确认 | 署名確認

Role	Name	Signature	Date
Survey Engineer			
Project Manager			
Owner Representative			
Electrical Engineer			
Structural Engineer			

9. Additional Notes | 补充说明 | 備考

NrgOpt — Enterprise Clean Energy Solutions / 企业清洁能源解决方案
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This checklist should be completed in full before proceeding to technical proposal stage.
本踏勘表应在进入技术方案阶段前完整填写。本チェックリストは技術提案段階に進む前に完全に記入してください。